

Simulation and Combined Spectral Analysis for In-Situ Soil Elemental Content Measurement

Abstract

Conventional methods for measuring soil carbon are often time-consuming, expensive, and require laboratory analysis. Neutron-gamma analysis (NGA) offers an alternative for rapid (practically in-situ) and non-destructive measurement of soil carbon. This paper describes mathematical methods to measure carbon levels in stimulated spectra of soil samples. Monte-Carlo simulations were carried out to create soil neutron stimulated gamma spectra using MCNP6.2 code. Two categories of analysis are investigated: peak fitting, and endmember fitting, a third method is introduced that combines the two into a single result. Evaluation of their effectiveness determines that the mixed model performed best overall.

Keywords: soil elemental content, gamma-ray spectroscopy, Pulsed Fast Thermal Neutron Analysis, Tagged Neutron Method, MCNP6.2 simulation, peak fitting, endmember fitting, non-destructive analysis, environmental monitoring, rapid measurement, agricultural soils

1. Introduction

Soil carbon is a key component of soil health and plays a crucial role in the global carbon cycle. Accurate measurement of carbon is essential for measuring soil quality, and its impact on the environment lal (2018) Traditional methods for measuring carbon are often time-consuming, expensive, and require laboratory analysis De Ros et al. (2025). In situ spectral analysis offers an alternative for rapid and non-destructive measurement of soil carbon Yakubova et al. (2019).

The Mobile Inelastic Neutron Scattering system (MINS) developed by the National soil dynamics laboratory USDA Agricultural Research Service (NSDL USDA-ARS) uses pulsed fast thermal neutron analysis (PFTNA) Yakubova et al. (2025) or the Tagged Neutron Method (TNM) Yakubova

et al. (2019) to measure soil carbon and other elements. Different experimental techniques are implemented in these methods. Regardless, both methods rely on analysis of gamma spectra providing information about the elemental composition of the soil. The analysis methods used to interpret the gamma-ray spectra are Peak Fitting for PFTNA and Endmember Fitting for TNM. This paper presents the combination of these methods in calibration. We simulate the MINS with experimental arrangements of equipment and its spectral acquisition using MCNP6.2 code. As samples, common soil minerals with varied carbon and moisture content. Then we apply different gamma-ray spectroscopy methods, to evaluate and compare their effectiveness in measuring carbon in soil.

1.1. State of the Art

INCLUDE A SECTION HERE ABOUT STATE OF THE ART

2. Physical Procedure and Simulation

The MINS system consists of a neutron generator, gamma-ray detectors, and a data acquisition system mounted on a mobile platform. The neutron generator emits fast neutrons that interact with the soil and the environment. In PFTNA, the detector measures during and between pulses of neutrons. This allows for the separate measurement of an inelastic scattering (INS) and thermo nuclear capture (TNC) spectra. In TNM, the neutron generator emits tagged neutrons, and the detectors are synchronized to capture gamma rays emitted from neutron interactions with the soil only.

2.1. Simulation in MCNP

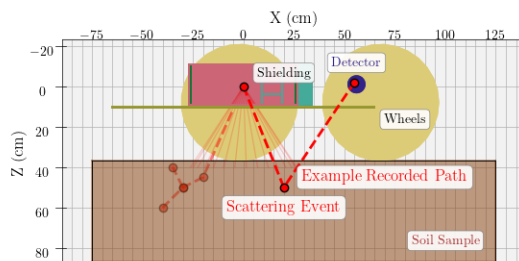


Figure 1: Side View Geometry of MINS Architecture

MCNP6.2 Werner et al. (2017) was used to simulate gamma-ray spectra resulting from neutron interaction with elements in soil samples. The
 54 resulting gamma rays that reach the detector is recorded. Each simulation
 56 modeled a soil sample undergoing neutron activation with a base soil mineral
 and varying concentrations of carbon and moisture content. The geometry
 58 was set up to mimic in situ measurement conditions, with a neutron source
 placed above a soil slab and a detector positioned to capture emitted gamma
 60 rays Kavetskiy et al. (2017). In Figure 1, the side view of the MCNP sim-
 ulation setup is shown, highlighting the arrangement of the neutron source,
 62 soil slab, and detector. The neutron generator produces neutrons in solid
 angle 0.4 sr directed down to soil (like in TNM modification of MINS). In
 64 this case, neutrons only interact with the soil, causing the emission of gamma
 rays characteristic of the elements present in the sample. The gamma-ray
 66 detectors capture these emissions, and the data acquisition system records
 the spectra for analysis.

68 In Figure 2, the top view of the MCNP simulation setup is presented, pro-
 viding a clear view of the spatial arrangement of the components, including
 70 the shielding around the neutron source and detector, as well as the wheels
 used for mobility in field applications.

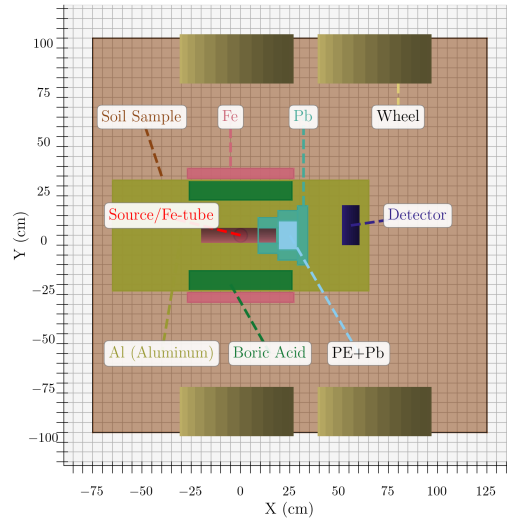


Figure 2: Cross section of MCNP simulation setup showing wheels and shielding (Top View)

- 72 Key simulation parameters include:
- 74 • **Neutron source energy 14.1 MeV:** API120 portable neutron (D-T generator) generator Kavetskiy et al. (2018)
 - 76 • **Soil slab dimensions:** 200 cm $\times$ 200 cm $\times$ 50 cm Wielopolski and Doron (2012)
 - **Detector type:** LaBr detector
 - 78 • **Tally:** F8 (pulse height tally) for gamma spectra

Table 1: Architectural Chemical Identifiers

Material	Element	ID	Weight %
PE+Pb	C	6000	4.2860
	H	1001	0.7140
	Pb	82000	95.0000
Pb	Pb	82000	100.0000
Boric Acid	H	1001	4.8535
	B10	5010	3.4981
	B11	5011	13.9923
	O	8016	77.6561
Al	Al	13027	100.0000
Fe-tube	C	006000.50c	0.0400
	Si	014000.60c	0.5000
	P	015031.50c	0.0230
	S	016000.60c	0.0150
	Cr	024000.50c	18.9998
	Mn	025055.60c	1.0000
	Fe	026000.50c	70.1723
	Ni	028000.50c	9.2499
Fe	Fe	26000	100.0000
Detector	Br79	35079	29.4600
	Br81	35081	30.6900
	La139	57139	34.8500
	Ce140	58140	5.0000
Wheel	H	1001	11.8371
	C	6000	88.1629

Table 1 lists the architectural chemicals and their identifiers for the soil
80 elements. Table 2 lists the identifiers for the soil elements.

Table 2: Soil Chemical Identifiers							
Element	C	H	O	Si	Na	Al	K
MCNP Identifier	6000	1001	8016	14000	11023	13027	19000

2.2. Common Soil Minerals

82 To investigate the effectiveness of gamma spectra analysis methods for soil
carbon measurement, we simulate common soil minerals, including Quartz,
84 Feldspar and Mica Sposito (2008). This data serves as a foundation for
applying various analysis techniques.

Table 3: The elemental weight composition of common soil minerals							
Material	C %	H %	O %	Si %	Na %	Al %	K %
Carbon	100.0	0.0	0.0	0.0	0.0	0.0	0.0
Water	0.0	11.2	88.8	0.0	0.0	0.0	0.0
Quartz	0.0	0.0	53.3	46.7	0.0	0.0	0.0
Feldspar	0.0	0.0	48.8	32.1	8.8	10.3	0.0
Mica	0.0	0.5	48.2	21.2	0.0	20.3	9.8

86 Table 3 presents the elemental weight composition of common soil miner-
als used in the simulations. To measure the effectiveness of analysis methods
88 for carbon measurement, we simulate combinations of soil minerals with vary-
ing carbon and water content. The simulations range from 0% to 15% carbon
90 (0%, 1%, ..., 15%) and from 0% to 20% (0%, 5%, ..., 20%) water content by
weight.

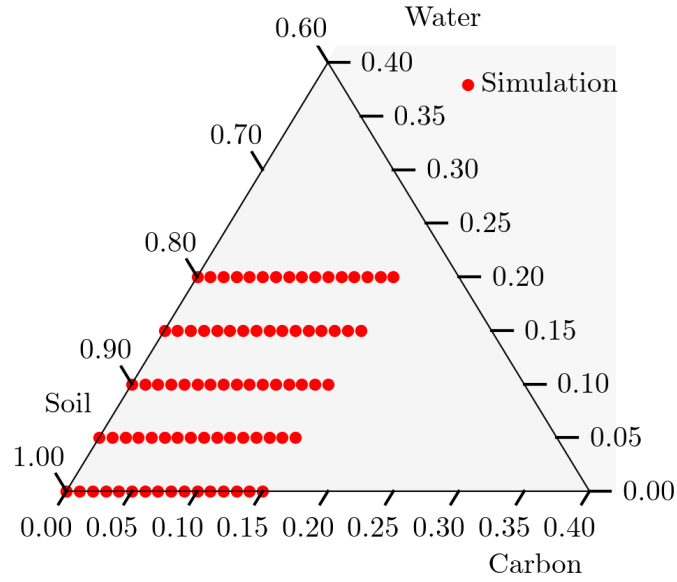


Figure 3: Simulated combinations of soil carbon and water levels

92 2.3. *Gamma Ray Spectra*

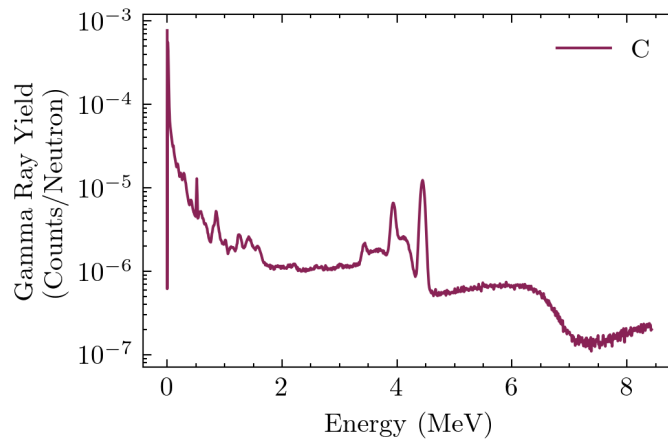


Figure 4: Example Simulated Carbon Spectrum

As shown in Figure 4, spectra are a representation of the energy distribu-
 94 tion of gamma rays captured by a detector. They are typically displayed as
 a graph with energy bins (or bands) on the x-axis and counts (or intensity)

96 per neutron (in simulation) on the y-axis. The observed spectrum $O(E)$ can
be represented by the integral Meng and Ramsden (2000):

$$O(E) = \int_0^\infty R(E, E_0) \cdot I(E_0) dE_0 \quad (1)$$

98 Here, $I(E_0)$ is the incident spectrum for all possible incident energies, and
 $R(E, E_0)$ is the detector response function, which accounts for the efficiency
100 of the detector.

This equation can be discretized as:

$$\begin{bmatrix} O_1 \\ \vdots \\ O_m \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1m} \\ \vdots & \vdots & & \vdots \\ R_{m1} & R_{m2} & \cdots & R_{mm} \end{bmatrix} \times \begin{bmatrix} I_1 \\ \vdots \\ I_m \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_m \end{bmatrix} \quad (2)$$

102 where m is the number of energy bins, ϵ_i ($i = 1, \dots, m$) is noise and R_{ij} is
the probability that an incident gamma-ray having an energy falling into bin
104 i will be detected in bin j .

2.4. Gaussian Energy Broadening as Detector Response

106 The F8 tally tracks the energy detected by the system, or incident spec-
tra. To model the deficiency in the detectors observed resolution, a response
108 function is applied. The response function used is Gaussian energy broaden-
ing (GEB) is a model for detector response. The effects of which can be seen
110 in Figure 5. It is applied in simulations by a random sampling of a Gaussian
distribution as such:

$$E' = E + \Delta E \quad (3)$$

112 where E is the true energy of the incoming particle, ΔE is a random vari-
able sampled from a Gaussian distribution defined by the full width at half
114 maximum (FWHM):

$$\text{FWHM} = a + b\sqrt{E} + cE^2, \quad (4)$$

where E is the energy of the particle. The units of a , b , and c are MeV,
116 $\text{MeV}^{1/2}$, and MeV^{-1} , respectively. The energy actually scored is sampled from
a Gaussian with that FWHM. The parameters a , b , and c were determined
118 experimentally as -0.026198 0.059551 -0.037176 respectively Werner et al.
(2017).

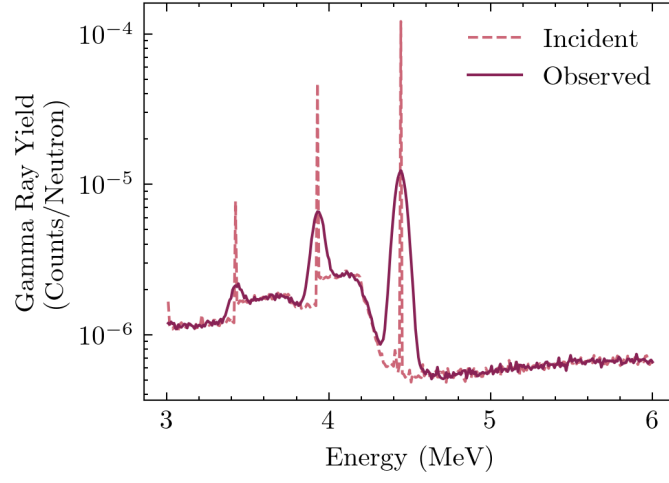


Figure 5: Response Function: Gaussian Energy Broadening (GEB)

120 2.5. Effects on Spectral Readings

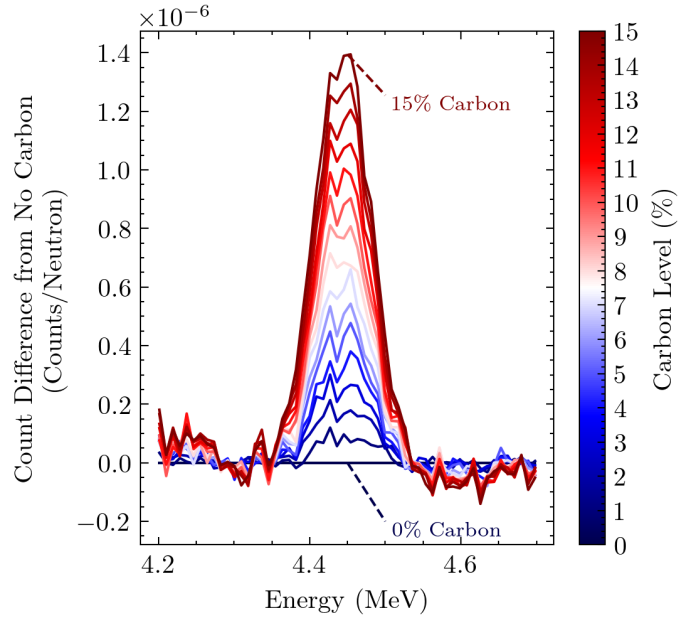


Figure 6: Carbon effect on Feldspar Spectrum

The spectral readings obtained from the MCNP simulations provide a
 122 detailed distribution of the gamma-ray emissions from the soil samples per

neutron, similar to what would be measured in a real-world scenario. Spectrums can be used to identify the presence of specific elements based on their characteristic energy peaks.

The effect of the carbon on the spectral readings can be seen near Carbons 4.44 MeV gamma peak in Figure 6. As the carbon content increases, the peak becomes more pronounced. Analysis methods can use this information to determine the carbon content of the soil.

As seen in Figure 7, peaks from Oxygen and Silicon interfere with the carbon peak, caused by gamma emissions from Oxygen and Silicon in the soil. This has an effect on the analysis that needs to be accounted for when determining the carbon content in the soil.

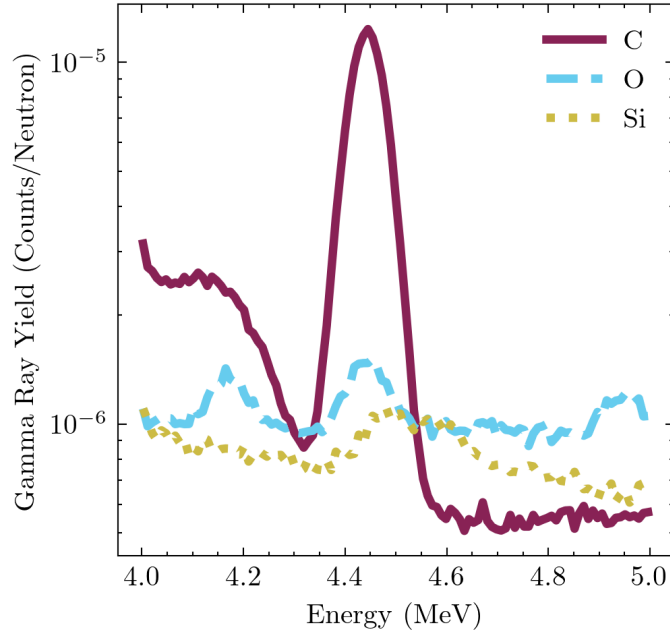


Figure 7: Carbon compared to Oxygen and Silicon gamma peaks

3. Data Analysis Methods

3.1. Curve Fitting

The curve fitting procedure begins by collecting data points (x_i, y_i) and selecting a suitable model function

Algorithm 1 Curve Fitting Procedure

Require: Data points (x_i, y_i) , model function $f(x, \text{params})$.

- 1: Initialize parameters params_0 .
 - 2: Define loss function: $L(\text{params}) = \sum_i (y_i - f(x_i, \text{params}))^2$.
 - 3: Use optimization algorithm to find $\text{params}_{\text{opt}}$ that minimize $L(\text{params})$.
 - 4: **return** Fitted parameters $\text{params}_{\text{opt}}$ and fitted curve $f(x, \text{params}_{\text{opt}})$.
-

138 $f(x, \text{params})$. Initial parameter values params_0 are chosen, and a loss func-
tion is defined as the sum of squared residuals: $L(\text{params}) = \sum_i (y_i -$
140 $f(x_i, \text{params}))^2$. An optimization algorithm is then employed to determine
the parameter set $\text{params}_{\text{opt}}$ that minimizes the loss function. The result is a
142 set of fitted parameters and the corresponding fitted curve $f(x, \text{params}_{\text{opt}})$.
This method is applied with the `scipy.optimize.curve_fit` function from
144 the scipy python package Virtanen et al. (2020) , which utilizes either the
Levenberg-Marquardt algorithm Moré (1978) (for unconstrained problems)
146 or the Trust Region Reflective (TRF) algorithm Branch et al. (1999) (for
constrained problems).

148 *3.2. Peak Baseline Fitting*

Algorithm 2 Peak Baseline Fitting Procedure

Require: Data points $\text{Spec}_i = (x_i, y_i)$ for bin i .

Require: Peak function $F_{\text{peak}}(x, \text{params}_{\text{peak}})$.

Require: Baseline function $F_{\text{baseline}}(x, \text{params}_{\text{baseline}})$.

Model function $F = F_{\text{peak}} + F_{\text{baseline}}$.

- 2: **for** each target peak **do**
Filter data points around the target peak.
 - 4: Initialize parameters params_0 .
Define loss function: $L(\text{params}) = \sum_i (y_i - F(x_i, \text{params}))^2$.
 - 6: Use TRF algorithm to minimize $L(\text{params})$ within bounds.
 $(x_i, Y_i) \leftarrow (x_i, F(x_i, \text{params}_{\text{opt}}))$.
 - 8: Subtract baseline: $Y_i \leftarrow Y_i - F_{\text{baseline}}(x_i, \text{params}_{\text{baseline, opt}})$.
Calculate peak area using Simpson's method: $\text{Area} = \text{simpson}(x_i, Y_i)$.
 - 10: **end for**
 - return** Peak Areas
-

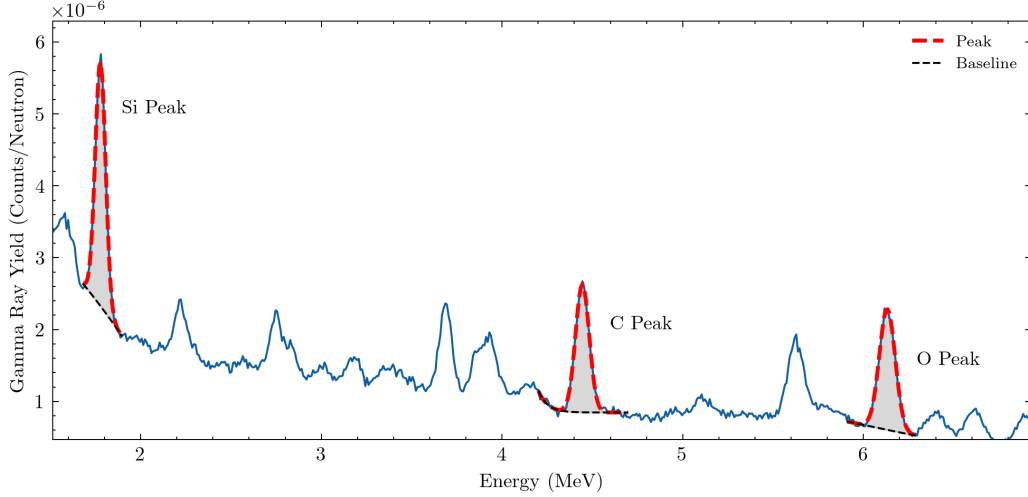


Figure 8: Peak Baseline fitting example for feldspar spectrum at 14% carbon level.

Peak fitting as shown in Figure 8 involves using the least-squares method
 150 in identifying and quantifying the baseline and peaks in the spectral data that
 correspond to specific soil components Gardner et al. (2011). This method
 152 is useful for extracting information about the concentration of individual
 elements in the soil. First, for each peak, the data is filtered to a local region
 154 around key peak areas, defined in Table 4.

Table 4: Energy Regions for Elemental Peak Analysis

Element	Energy Region (MeV)
C	[4.2, 4.7]
Si	[1.675, 1.9]
O	[5.9, 6.3]

The fitting function is defined as:

$$F = F_{\text{peak}} + F_{\text{baseline}} \quad (5)$$

156 where F_{peak} is the peak function (e.g., Gaussian) and F_{baseline} is the base-
 line function (e.g., linear or exponential falloff).

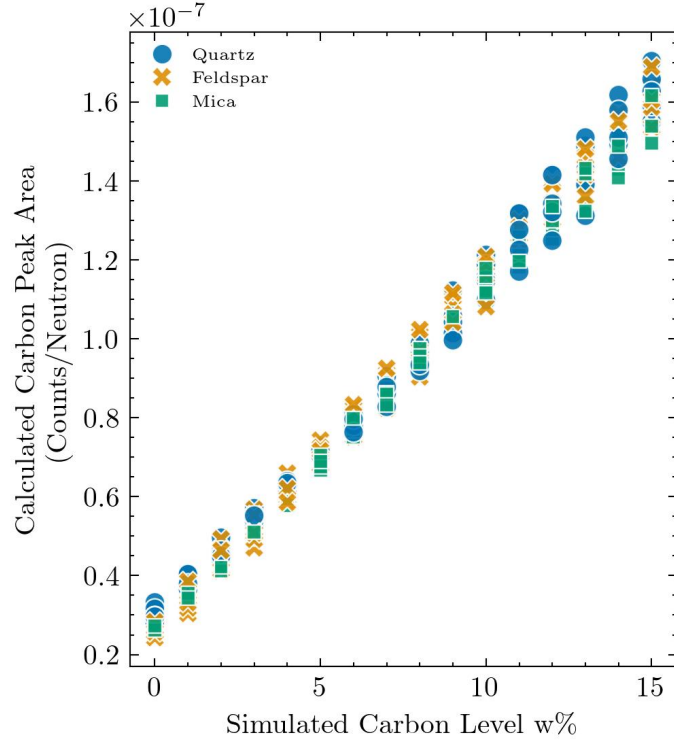


Figure 9: Peak areas are correlated with soil carbon levels over all mineral/carbon/water combinations

Table 5: Function parameterizations for peak fitting

Function Type	Example Expression
Peak	
Gaussian	$a \cdot \exp(-((x - b)^2)/c^2) + d$
Baseline	
Linear	$ax + b$
Exponential Falloff	$b \cdot \exp(-c \cdot (x - a)) + d$

158 F is fitted to the data using the Trust Region Reflective (TRF) algorithm
Branch et al. (1999). The specific functions for peaks and baselines used are
160 shown in Table 5. Initial parameters and bounds are defined per function
and spectra analyzed in Table 6 and Table 7. This begins the fitting process

162 with assumptions about the peak position and shape. For example, the a
 164 term in the gaussian function, which controls amplitude, is restricted to be
 positive but not taller than the peak presented.

Table 6: Initial Guess Parameterization for Fitting Functions

Function	Initial Guess $p_0 = [a, b, c, \dots]$
Linear	$[0, \min(y)]$
Gaussian	$[0.5 \cdot (\max(y) - \min(y)), \text{mean}(x), \frac{x_{\max} - x_{\min}}{6}]$
Exponential Falloff	$[\min(x), 0.5 \cdot (\max(y) - \min(y)), 1, \min(y)]$

Table 7: Parameter Bounds for Fitting Functions

Function	Lower Bounds	Upper Bounds
Linear	$[-10^9, 0]$	$[0, \max(y)]$
Gaussian	$[0, x_{\min}, \frac{x_{\max} - x_{\min}}{100}]$	$[\max(y) - \min(y), x_{\max}, \frac{x_{\max} - x_{\min}}{2}]$
Exponential Falloff	$([-\infty, -\infty, -\infty, -\infty])$	$([\infty, \infty, \infty, \infty])$

After curve fitting, the baseline function is subtracted from the fitted func-
 166 tion to isolate the peak, and the area under the peak is calculated using the
 Simpsons method implemented by `scipy.integrate.simpson` Cartwright
 168 (2017) to quantify the concentration of the corresponding element or com-
 pound in the soil. As shown in Figure 9, the peak areas are correlated with
 170 the carbon levels in the soil samples. The variance in peak area between sam-
 ples with similar carbon levels may be attributed to other soil components
 172 affecting the peak fitting. This includes elements such as oxygen and silicon,
 the ladder of which, as in Figure 10, is also shown to have a non-negligible
 174 correlation with the carbon peak area.

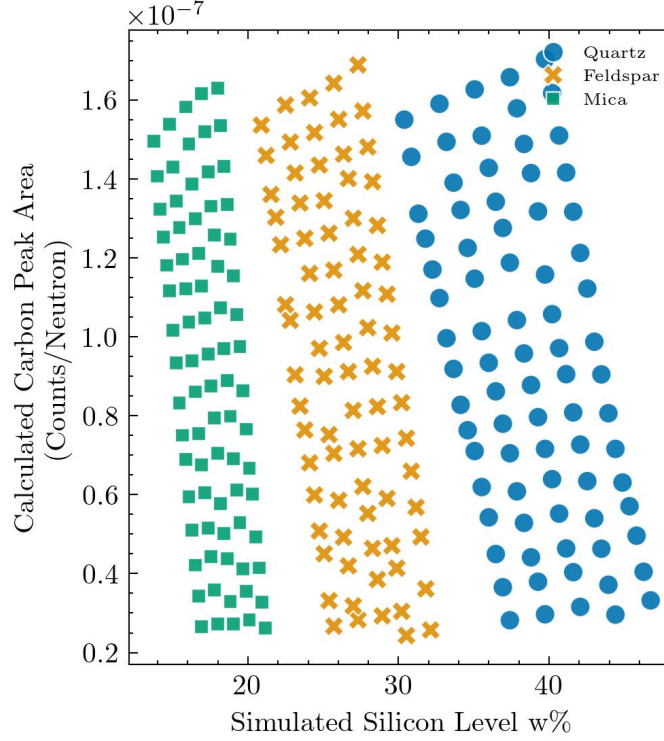


Figure 10: Silicon levels are correlated with carbon peak areas over all mineral/carbon/water combinations

3.3. Spectral Mixture of Endmembers

176 Spectral results can be modeled as a mixture of several component spec-
 178 tra, known as endmembers. This approach is useful in scenarios where the
 observed spectrum is a mixture of distributions. The linear spectral mixture
 model Adams et al. (1995) can be expressed as:

$$O(E)_i = \sum_{k=1}^n t_{ik} O_{ik} + \epsilon_i \quad \text{and} \quad \sum_{k=1}^n t_{ik} = 1, \quad (6)$$

180 Where $O(E)_i$ is the intensity in bin i and can be represented as a weighted
 sum of the intensities from each endmember O_{ik} , with coefficients given by
 182 the fractions t_{ik} . The term ϵ_i accounts for error.

3.4. Spectral Mixture Fitting of Endmembers

184 The Spectral Mixture Fitting of Endmembers leverages the Spectral Mix-
 ture of Endmembers model Eq 6 to fit known spectral components to the

Algorithm 3 Endmember Fitting Procedure

Require: Target $\text{Spec}_{\text{target},i} = (x_i, y_i)$.

Require: Endmember $\text{Spec}_{k,i} = (x_i, O_{k,i})$.

Filter data.

Initialize parameters $A_{k,0} = [1/k, \dots, 1/k]$.

3: Define loss function: $L(A) = \sum_i (y_i - \sum_k A_k \cdot O_{k,i})^2$.

Use TRF algorithm to minimize $L(A)$ bounded within $[0, 1]$.

return The final parameters: $A_{k,\text{opt}}$

186 observed spectrum. This was applied by Bishnoi et al. Bishnoi et al. (2019)
using 3 endmembers. The goal is to express the observed spectrum as a
188 linear combination of several known component spectra (endmembers), each
representing a specific soil constituent. To do this, we set $t_{ik} = A_k$, where
190 A_k are the coefficients representing the concentration of each component in
the mixture.

192 The combined spectral function is defined as:

$$F_{c_i} = \sum_k A_k \cdot O_{k,i} = A_C \cdot O_{C_i} + \dots + A_{Na} \cdot O_{Na_i} \quad (7)$$

where F_{c_i} is the combined spectral function at bin i , and $O_{k,i}$ are the ob-
194 served spectral functions of individual components at bin i . In Figure 11, we
illustrate the result of the fitting process. Beginning with the spectrum with
196 the highest coefficient, we stack plot the linear combination of the component
spectra weighted by their respective coefficients A_k . The fitting is performed
198 using the curve fitting procedure outlined earlier, F_c does not match its ob-
served spectrum exactly. This is further discussed in the conclusion.

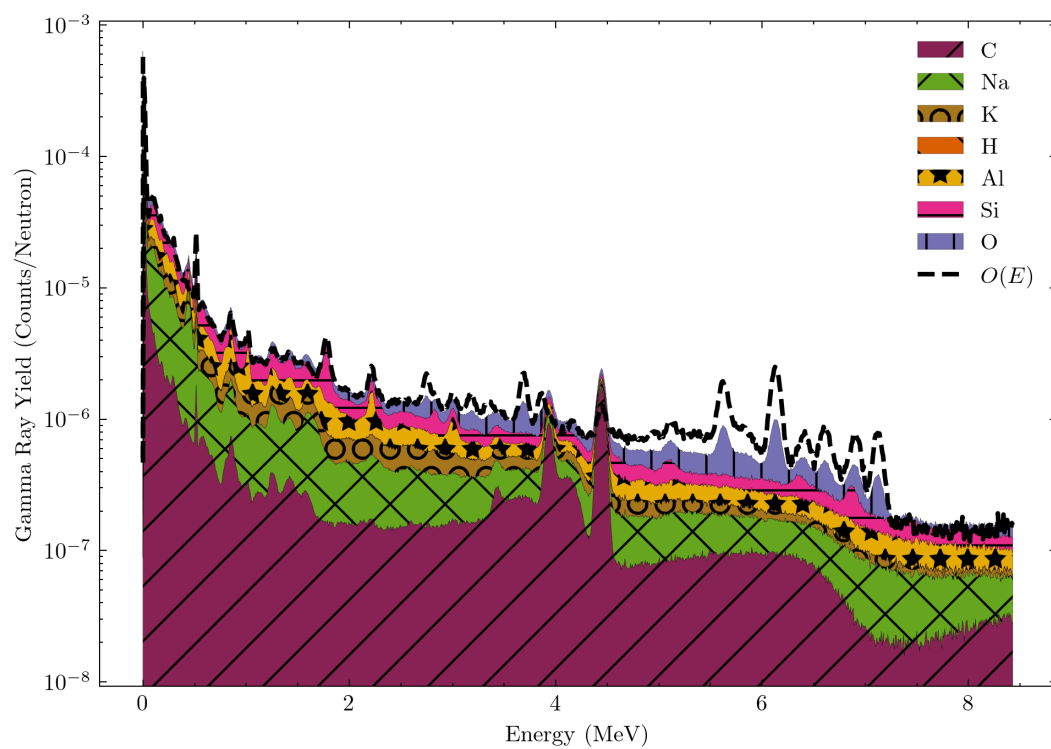


Figure 11: Stack plot of endmember fitting result for feldspar spectrum at 2% carbon 10% water level.

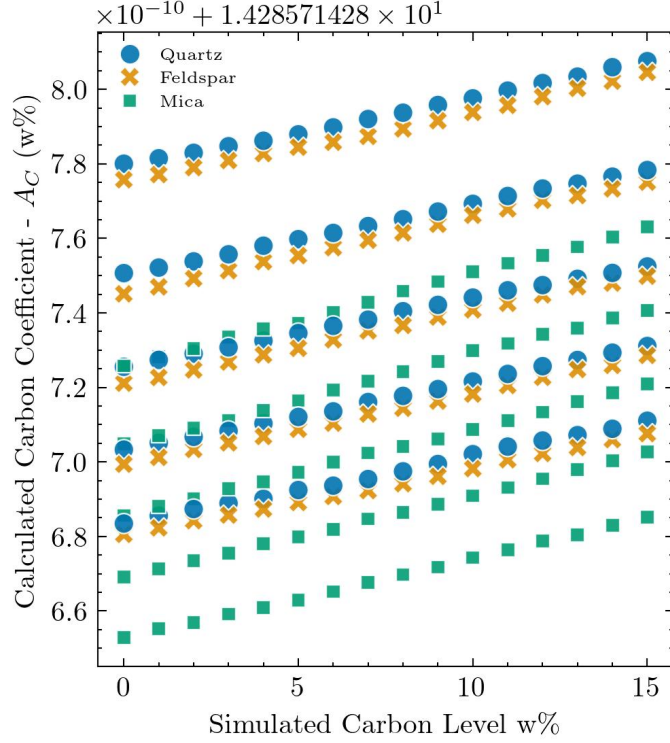


Figure 12: Carbon Coefficient is correlated with soil carbon levels over all mineral/carbon/water combinations

200 The carbon coefficient A_C is then used for calibration against known
 202 carbon concentrations to predict the carbon content in unknown samples.
 204 We can see by Figure 12 that there is a positive correlation between the
 206 calculated carbon coefficients from the SMFE method and the actual carbon
 levels in the soil samples. Unlike in peak fitting, there is a greater spread of
 predicted carbon coefficients for similar carbon levels. This is because the
 effect of silicone and oxygen has a greater effect on the methods calculation.

3.5. Multivariate Calibration Using Singular Value Decomposition

208 We employ a multivariate calibration technique. We extract multiple fea-
 210 tures from the spectral data, such as peak area via Peak Baseline Fitting and
 the carbon coefficient from endmember fitting. The data can be represented
 in matrix form as:

$$G = \begin{bmatrix} g_{11} & g_{12} & \cdots & g_{1n} \\ g_{21} & g_{22} & \cdots & g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ g_{m1} & g_{m2} & \cdots & g_{mn} \end{bmatrix}$$

212 for soil sample m and augmented feature n (peak area, endmember coefficients). A bias term is also included in the model to account for any constant
214 offset in the data.

The relationship between these features and the carbon content can be
216 modeled as:

$$y = Gx + \epsilon$$

where y is the vector of observed carbon content, x is the vector of coefficients
218 to be determined, and ϵ represents the error term.

To calibrate our model, we employ Singular Value Decomposition (SVD)
220 to analyze the relationship between the features found in analysis and the carbon content.

222 A matrix G , can be decomposed using SVD as follows:

$$G = U\Sigma V^T \quad (8)$$

Using this decomposition, we can find the SVD solution to least squares,
224 or convex optimization problems. This allows us to determine the coefficients that best fit our model to the observed data.

226 To solve for the coefficients x in the equation $Gx = b$, where b is the observed data, we can use the pseudoinverse of G :

$$x = G^+b \quad (9)$$

228 where G^+ is the pseudoinverse of G . The pseudoinverse can be computed from the SVD components as:

$$G^+ = V\Sigma^+U^T \quad (10)$$

230 where Σ^+ is obtained by taking the reciprocal of the non-zero singular values in Σ and transposing the resulting matrix.

232 For example, if Σ is a diagonal matrix with singular values $\sigma_1, \sigma_2, \dots, \sigma_r$
 (and possibly zeros for the remaining entries), then Σ^+ is given by:

$$\Sigma^+ = \begin{bmatrix} \frac{1}{\sigma_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\sigma_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\sigma_r} \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} \quad (11)$$

234 where the reciprocals are taken only for the nonzero singular values.

In our implementation, we utilize the `numpy.linalg.svd` function from
 236 the NumPy Harris et al. (2020) Python package to perform the singular value
 decomposition.

238 The final coefficients obtained from the SVD calibration are then used to
 predict the carbon content based on the extracted features from introduced
 240 spectral data:

$$\hat{y} = G_{test}x \quad (12)$$

where $\hat{y}$ is the predicted carbon content.

242 4. Comparison and Results

4.1. Metrics for Model Evaluation

244 To compare the performance of different models, we split the data ran-
 domly between 70% training and 30% testing sets. We vary the combination
 246 of methods used, then calibrate the results of those methods to the carbon in
 the testing set using SVD. We use the Mean Squared Error (MSE) between
 248 the defined and predicted carbon values. We define the error for each tested
 sample i as:

$$\text{Error}_i = (y_i - \hat{y}_i)^2 \quad (13)$$

250 where y_i is the simulated carbon content and $\hat{y}_i$ is the predicted carbon
 content for sample i in the testing set. The Mean Squared Error (MSE) is
 252 then calculated as:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \text{Error}_i \quad (14)$$

where N is the total number of tested samples.

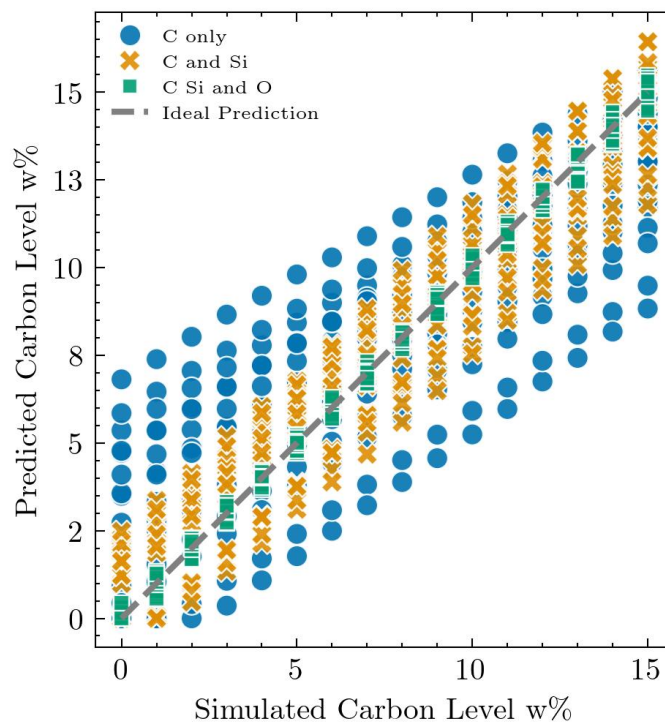


Figure 13: Multivariate calibration improves accuracy

The effects of multivariate analysis can be seen most clearly with the
 256 Edgemember Fitting method as in Figure 13. By including additional vari-
 258 ables such as silicon and oxygen coefficients alongside carbon, the model can
 260 better account for overlapping spectral features and interferences, and the
 262 predictions become more accurate. The most accurate of both methods was
 the multivariate analysis including carbon, silicon, and oxygen endmember
 coefficients and peak areas as variables respectively as seen in Table 8. Com-
 bining both the peak areas and the component fitting coefficients into a single
 model improved the accuracy further.

Table 8: Mean Squared Error (MSE) Comparison of Different Calibration Methods

Method	MSE		
<i>Variable Element:</i>	<i>C, Si, O</i>	<i>C, Si</i>	<i>C</i>
Combined	1.98e-06		
Edgemember Coeff.	4.28e-06	2.11e-4	7.52e-4
Peak Fitting	1.00e-05	1.37e-05	1.79e-05

264 4.3. Conclusion and Discussion

266 This study demonstrates the use of multi-variate calibration to enhance soil carbon content predictions by integrating features from various analysis methods.

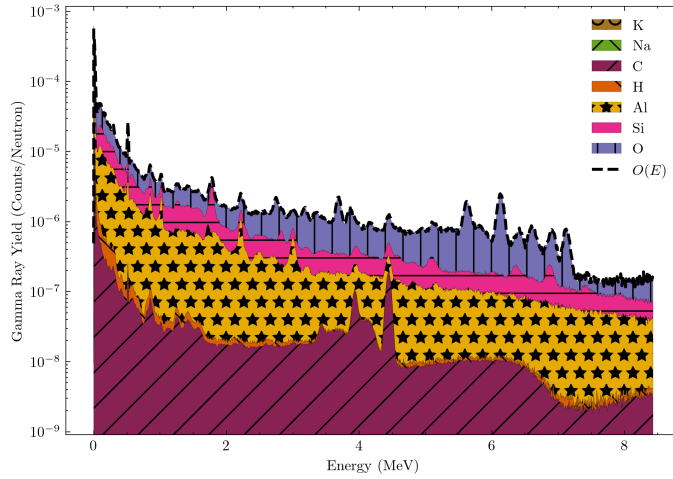


Figure 14: Edgemember Fitting (MSRE Error Metric)

268 In Figure 11 we observe that the fitted coefficients do not sum exactly
 270 to the observed spectrum. Furthermore, the final coefficients are close to
 272 their initial values. This is because the curve fitting error is based on MSE
 minimization, which is more sensitive to larger values found in lower energy
 regions. As shown in Figure 14 and Table 9 MSRE (Mean Squared Relative
 Error) improves this.

Table 9: Comparison of coefficients using MSRE fitting

Element	Coefficient	True
C	0.0156	0.0200
H	0.1616	0.0156
O	0.4777	0.5130
Si	0.1769	0.1862
Na	0.0000	0.0000
Al	0.1681	0.1788
K	0.0000	0.0864

274 As seen in Table 10 in the Univariate case, this results in a lower prediction
error, but in the Multivariate case, the error of the predictions remains fixed,
276 while the MSE version improves to be better than its counterpart.

Table 10: Comparison of Different Edgemember Fitting Error Metrics

Variables	MSE Fitted	MSRE Fitted
C Si and O	4.28e-06	1.47e-05
C and Si	2.11e-04	2.28e-05
C only	7.52e-04	2.29e-05

Our results demonstrate a significant performance increase, showing that
278 this combined approach yields more accurate predictions of soil carbon con-
tent than either method used alone. However, there are many soil types, the
280 soil minerals listed by Sposito Sposito (2008) are only one part of the possible
makeups that the MINS would encounter in the field. The simulations pre-
282 sented also do not include variance in density. More detailed simulations and
spectra to match PFTNA and TNM methods would help to further validate
284 the results presented here.

The calibration step can further include in its calculations the results of
286 various methods. As the dataset grows, more methods of analysis such as
convolutional neural networks as presented in Kim et al. Kim et al. (2025).
288 Other models, such as deep learning models, or other deconvolution methods
can be ensembled together to improve accuracy further.

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